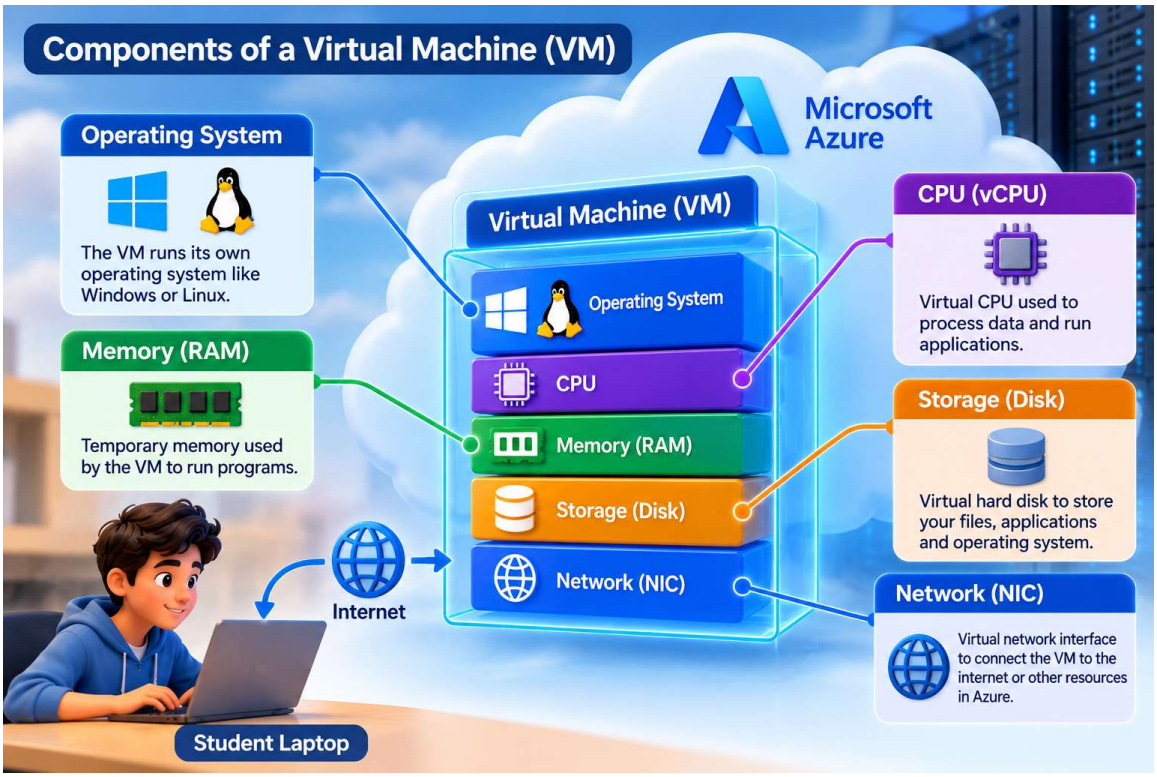
Virtual Machine (VM) in Microsoft Azure

1. What is a Virtual Machine?

A Virtual Machine (VM) is a computer created using software. It works like a physical computer and can have its own operating system, CPU, memory, storage, and network connection.

Microsoft Azure lets you create and use a VM hosted in Microsoft's data centre and connect to it over the internet. You do not need to buy another physical computer to get started.



2. A Simple Story

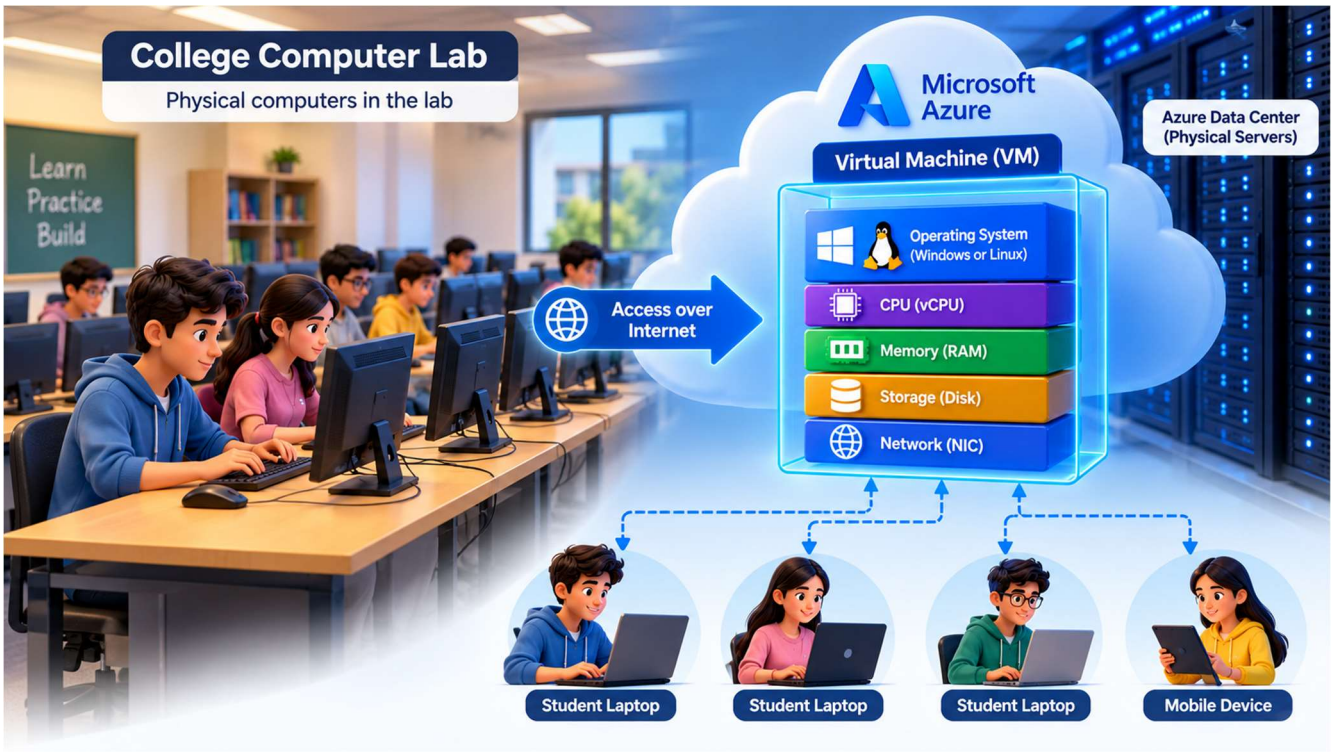
1. Imagine you already have a Windows laptop with 8 GB RAM.
2. Your teacher asks you to practise Linux or run software that needs more memory.
3. You do not want to buy another computer or change the operating system on your laptop.
4. You create a Virtual Machine in Microsoft Azure and choose the operating system and size you need.
5. You connect to the VM over the internet and practise on that separate computer.

3. Why Use Azure if We Already Have a Laptop?

- **Different operating systems:** Use a Linux VM even if your personal laptop runs Windows.
- **More computing power:** Choose a VM with more CPU or RAM when your laptop is not powerful enough.
- **Access from different places:** Connect to your VM from home, college, or another location using the internet.

- **Safe practice:** Experiment with software and server settings in a separate environment.
- **No need to own a server:** Azure provides the underlying physical servers in its data centres.

4. 3D Illustration: College Lab and Azure VM



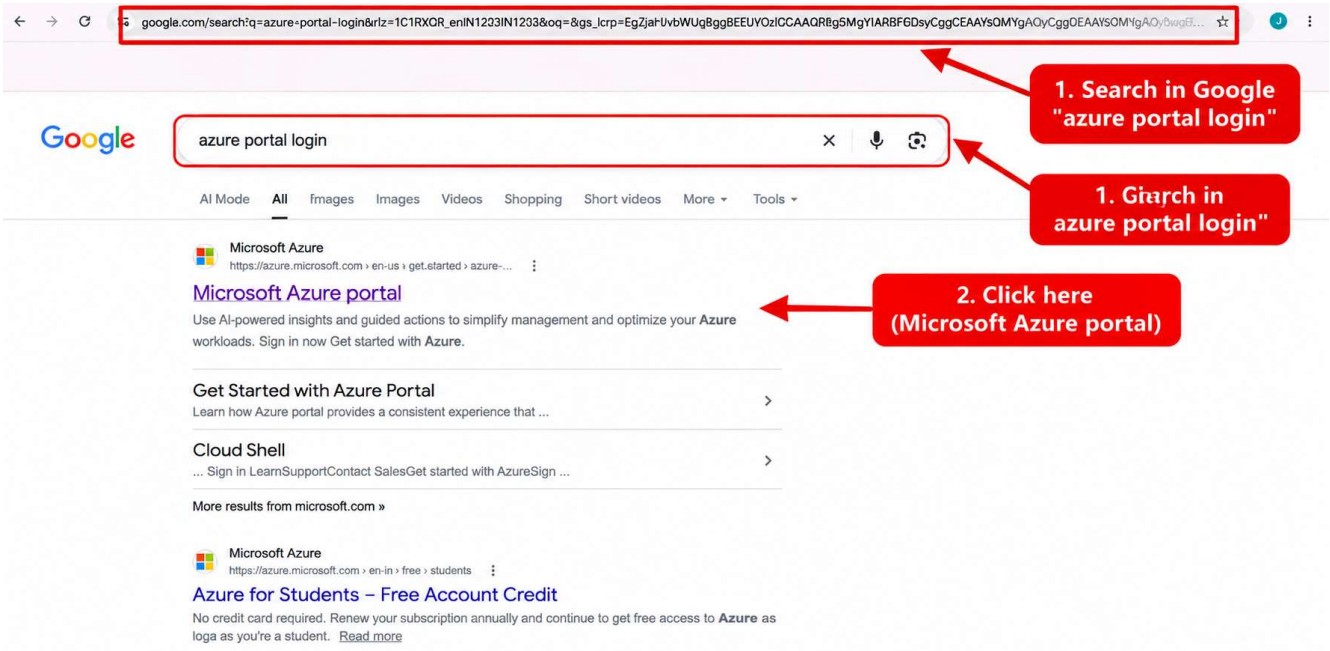
The college computers connect through the internet to a Virtual Machine hosted on Azure infrastructure.

Create and Configure an Azure Virtual Machine

Azure Portal Login – Screenshot Guide

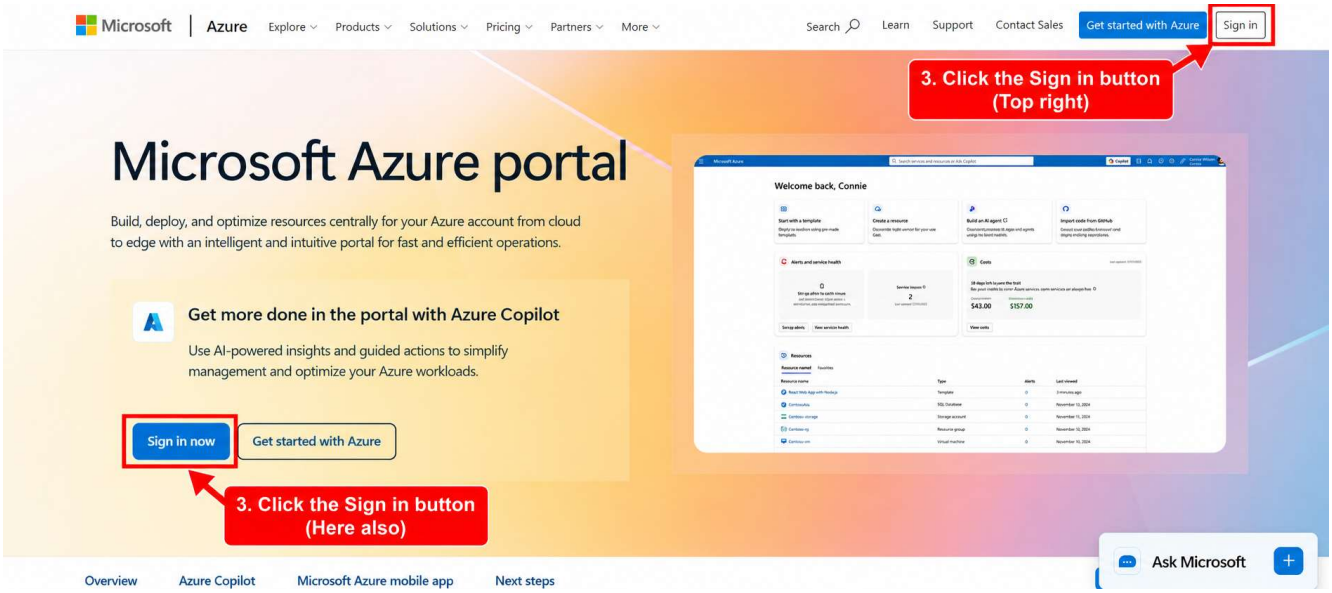
Step 1: Use the browser address bar to navigate/search for Azure portal login.

Step 2: Select the official “Microsoft Azure portal” result.


google.com/search?q=azure+portal+login&rlz=1C1RXQR_enIN1203IN1203&oq=&gs_lcrp=EgZjaHUvbWUgBggBEEUYOzICCAAQRBgSMgYIARBFGDsyCggCEAAYsOMYgAOyCggDEAAYsOMYgAOyCggEEAAYsOMYgAOyCggFEAAYsOMYgAOyCggGEAAYsOMYgAOyCggHEAAYsOMYgAOyCggIEAAYsOMYgAOyCggJEAAYsOMYgAOyCggKEAAYsOMYgAOyCggLEAAYsOMYgAOyCggMEAAYsOMYgAOyCggNEAAYsOMYgAOyCggOEAAYsOMYgAOyCggPEAAYsOMYgAOyCggQEAAYsOMYgAOyCggREAAYsOMYgAOyCggSEAAYsOMYgAOyCggTEAAYsOMYgAOyCggUEAAYsOMYgAOyCggVEAAYsOMYgAOyCggWEAAYsOMYgAOyCggXEAAYsOMYgAOyCggYEAAYsOMYgAOyCggZEAAYsOMYgAOyCgoaEAAYsOMYgAOyCgobEAAYsOMYgAOyCgocEAAYsOMYgAOyCgodEAAYsOMYgAOyCgoeEAAYsOMYgAOyCgofEAAYsOMYgAOyCggAEAAYsOMYgAOyCggBEAAYsOMYgAO

Step 3: Sign In to Microsoft Azure

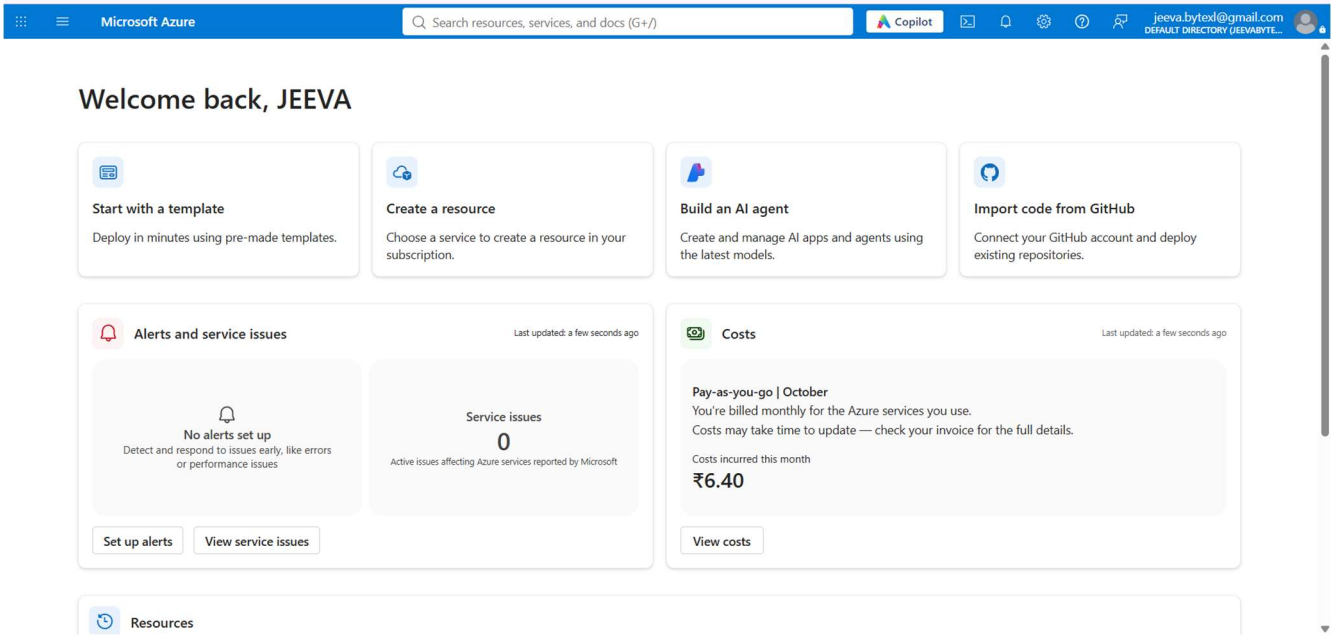
On the Microsoft Azure page, click either of the highlighted Sign in buttons to continue.



Both the top-right Sign in button and the Sign in now button in the main section are highlighted in red.

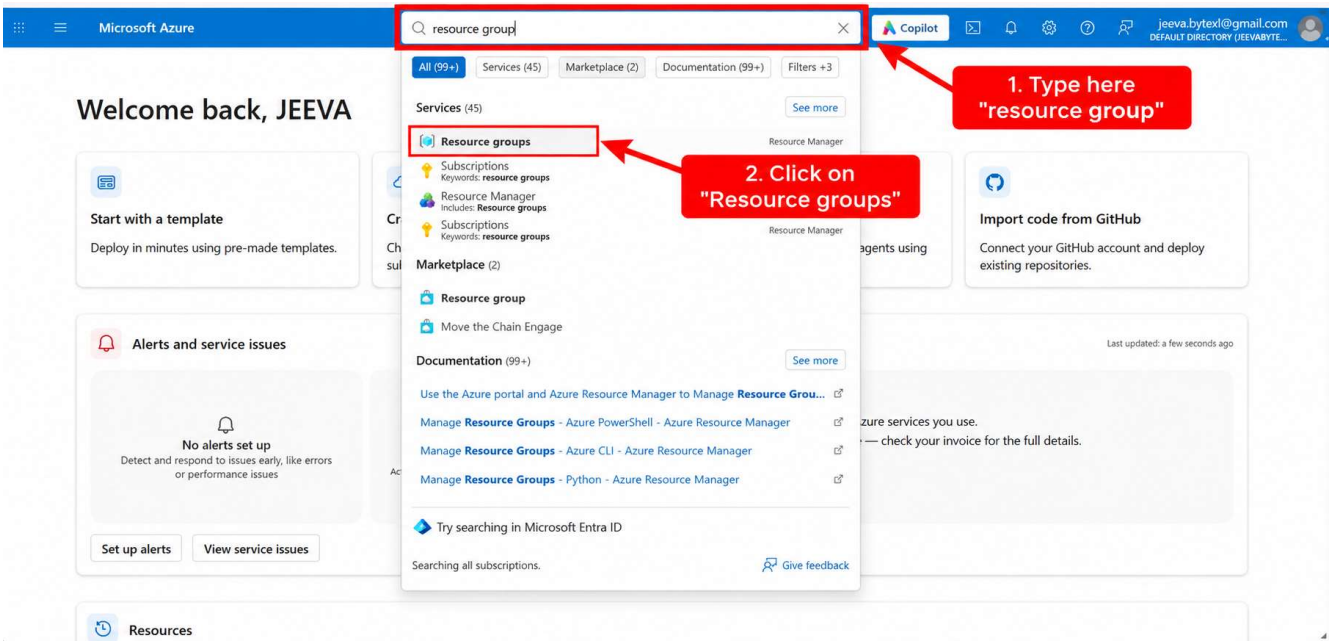
Step 4: Azure Portal Dashboard

After signing in successfully, the Azure portal dashboard appears. From this page, you can access Azure services and create resources such as Virtual Machines.



Step 5: Find Resource Groups in the Azure Portal

From the Azure dashboard, type “resource group” in the top search bar, then select “Resource groups” from the Services results.



What Is an Azure Resource Group?

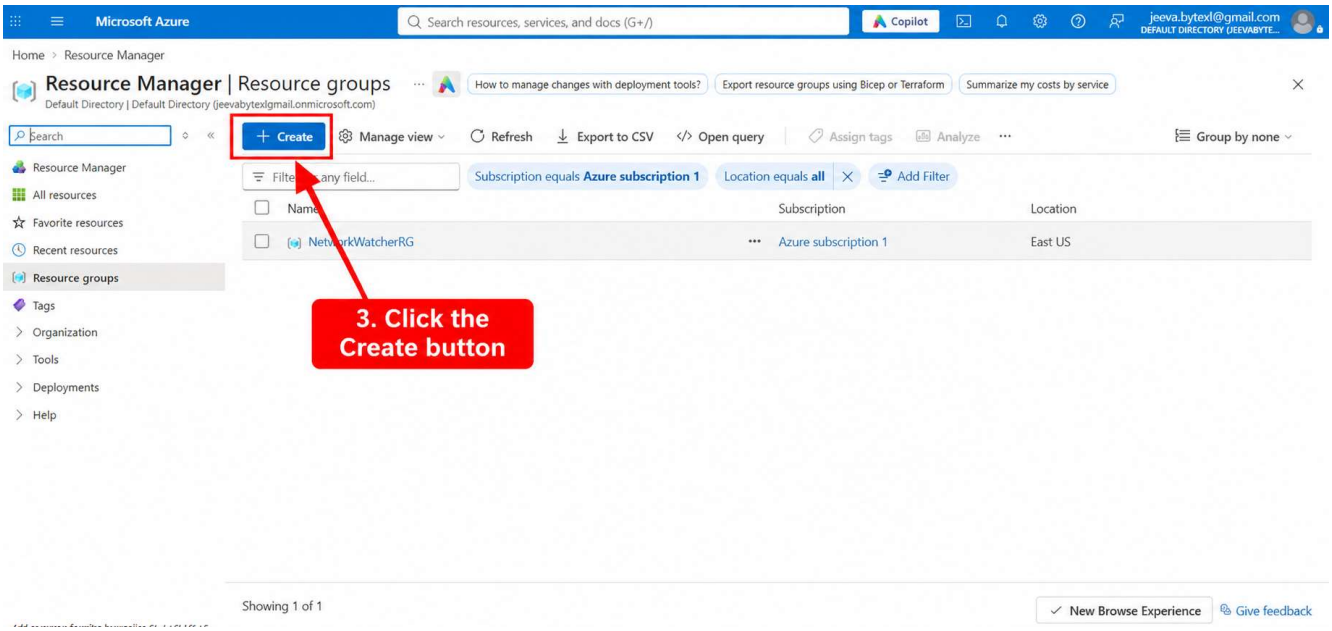
A Resource Group in Microsoft Azure is a logical container used to organize and manage related Azure resources, such as Virtual Machines, storage accounts, and virtual networks.



Think of a resource group as a folder that keeps related Azure resources together.

Step 6: Create the Resource Group

On the Resource groups page, click the + Create button to begin creating a new resource group.



The red box and arrow show where to click.

Step 7: Understand the Resource Group Fields

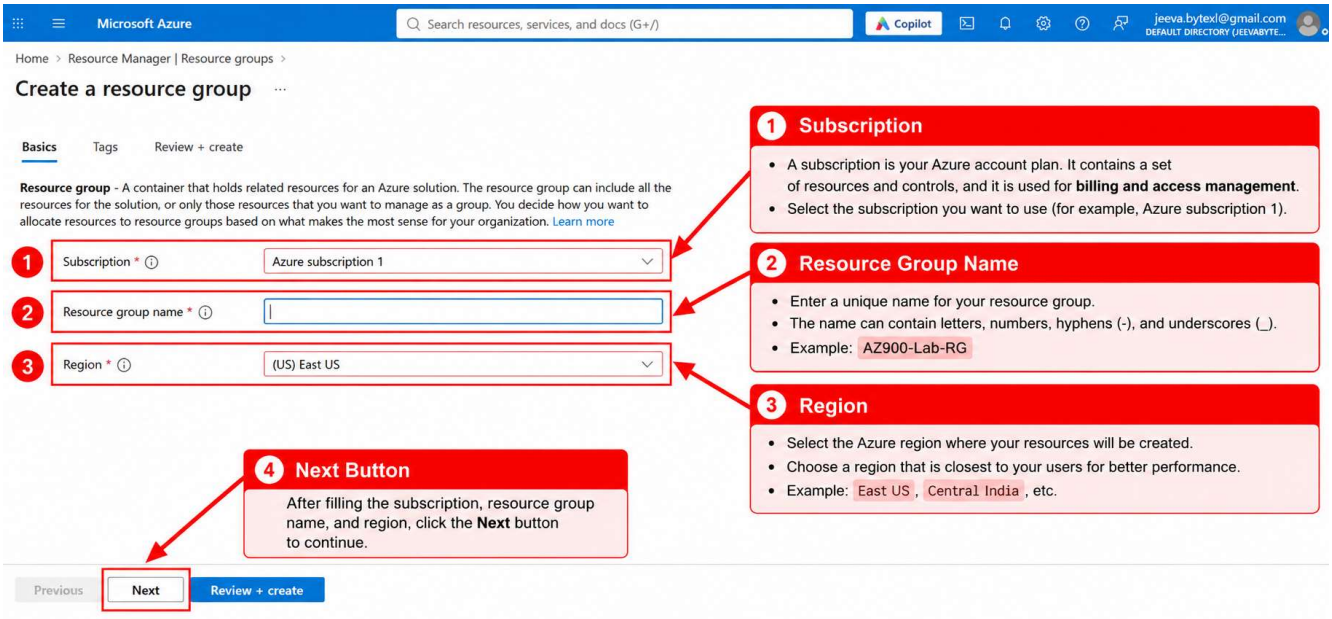
Before clicking Next, understand the three required fields on the Basics tab.

1. Subscription: An Azure subscription is used to manage resource usage, billing, and access. Select the subscription you want to use.

2. Resource Group Name: A unique name used to identify the container that organizes related Azure resources. Example: AZ900-Lab-RG.

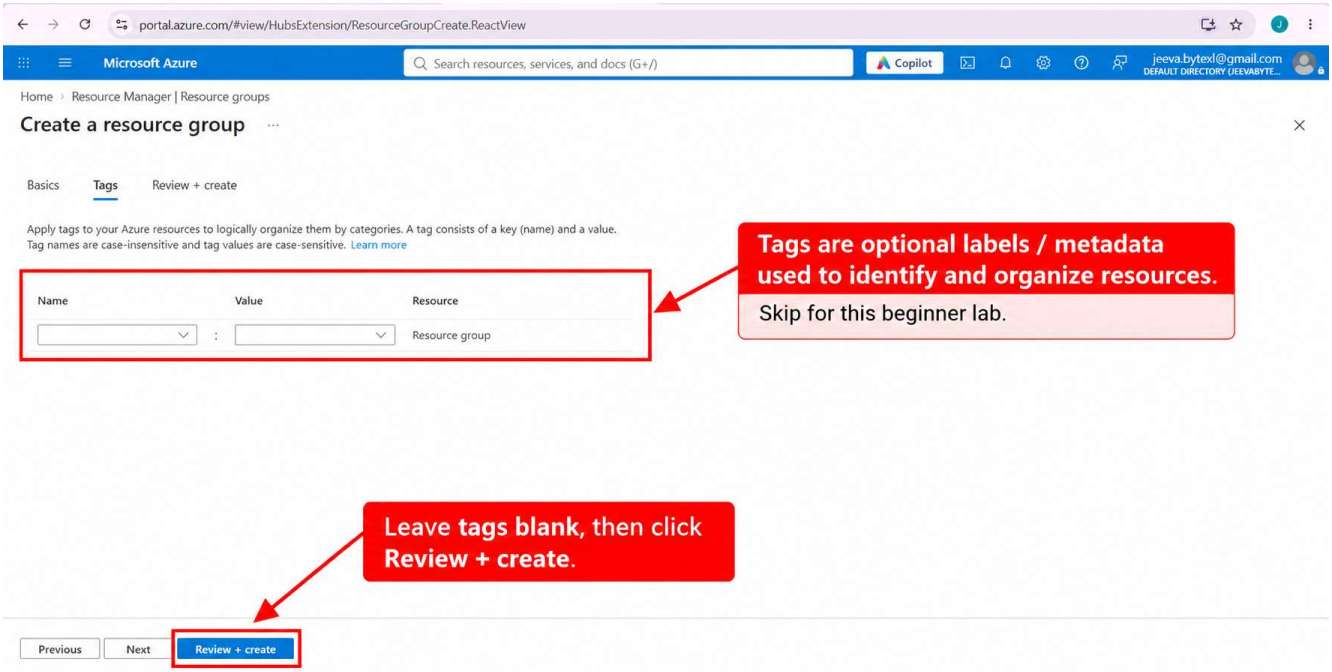
3. Region: The geographical location associated with the resource group and the location where you can deploy resources. Choose a suitable region, such as East US or Central India.

Easy way to remember: Subscription = account and billing; Resource Group Name = folder name; Region = geographical location.



Step 8: Tags – Optional Labels or Metadata

Tags are labels (metadata) that help identify, categorize, and organize Azure resources. For this beginner lab, leave the Name and Value fields blank because tags are optional.



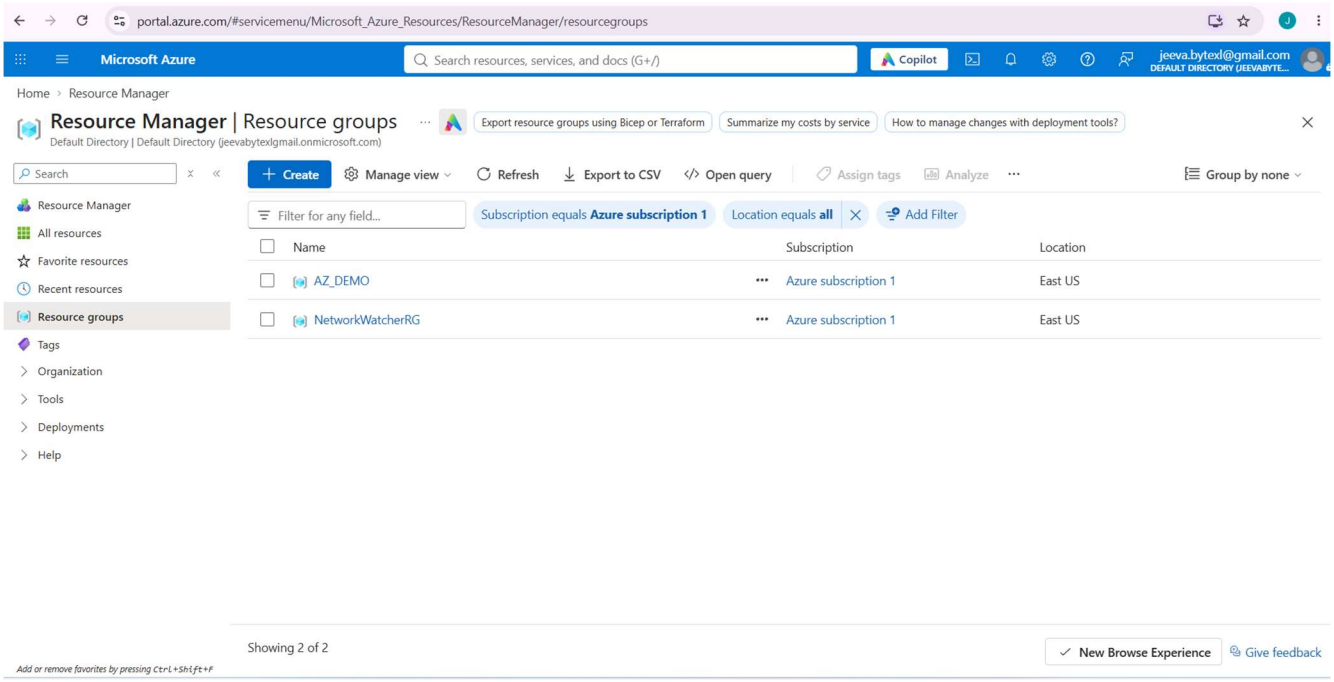
Step 9: Review + Create

Click Review + create to check the resource group settings before creating it. On the review page, confirm the subscription, resource group name, and region. If validation succeeds and the details are correct, click Create to create the resource group.

After deployment completes, open Resource groups and check that your new resource group appears in the list. You can then create the Virtual Machine inside this resource group.

Step 10: Refresh and Verify the Resource Group

After creating the resource group, return to the Resource groups page and click Refresh. Check the Name column to confirm that the new resource group appears in the list.



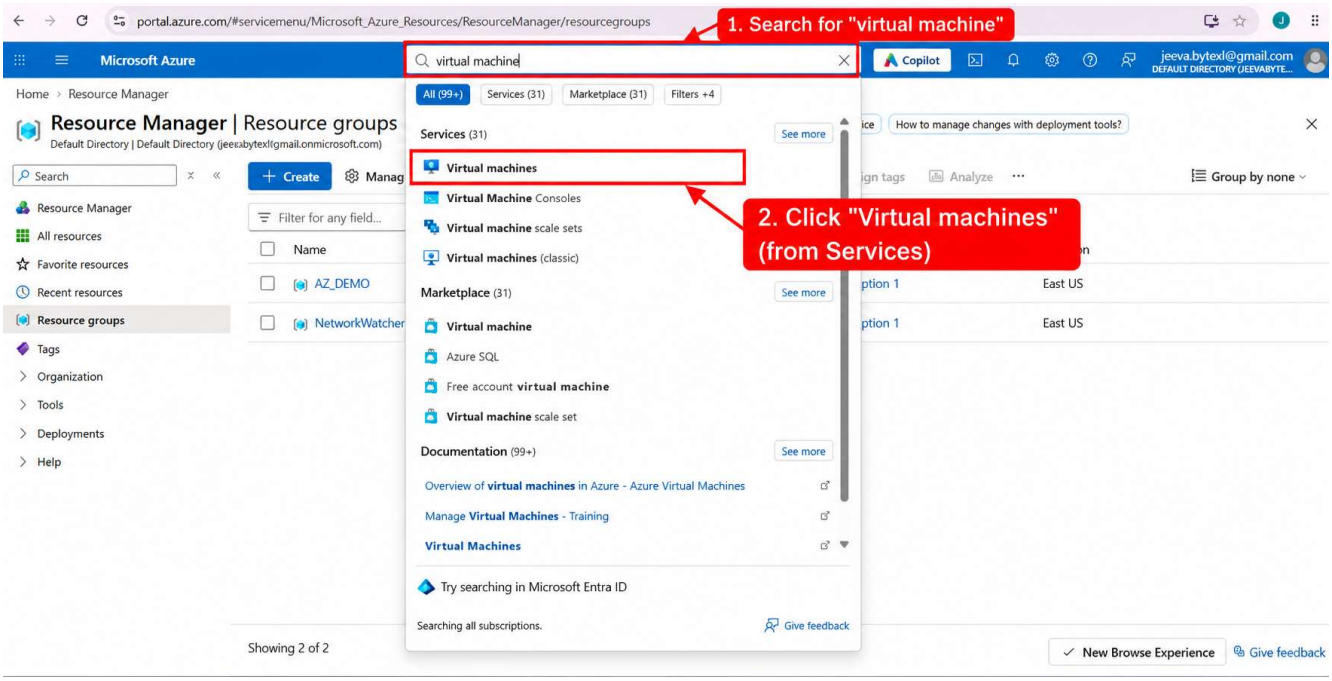
If the resource group does not appear, confirm that creation completed successfully and check the name you entered.

Next Step

Once the resource group is visible, you can begin creating a Virtual Machine inside it. Check estimated costs before deploying resources.

Step 11: Search for Virtual Machines

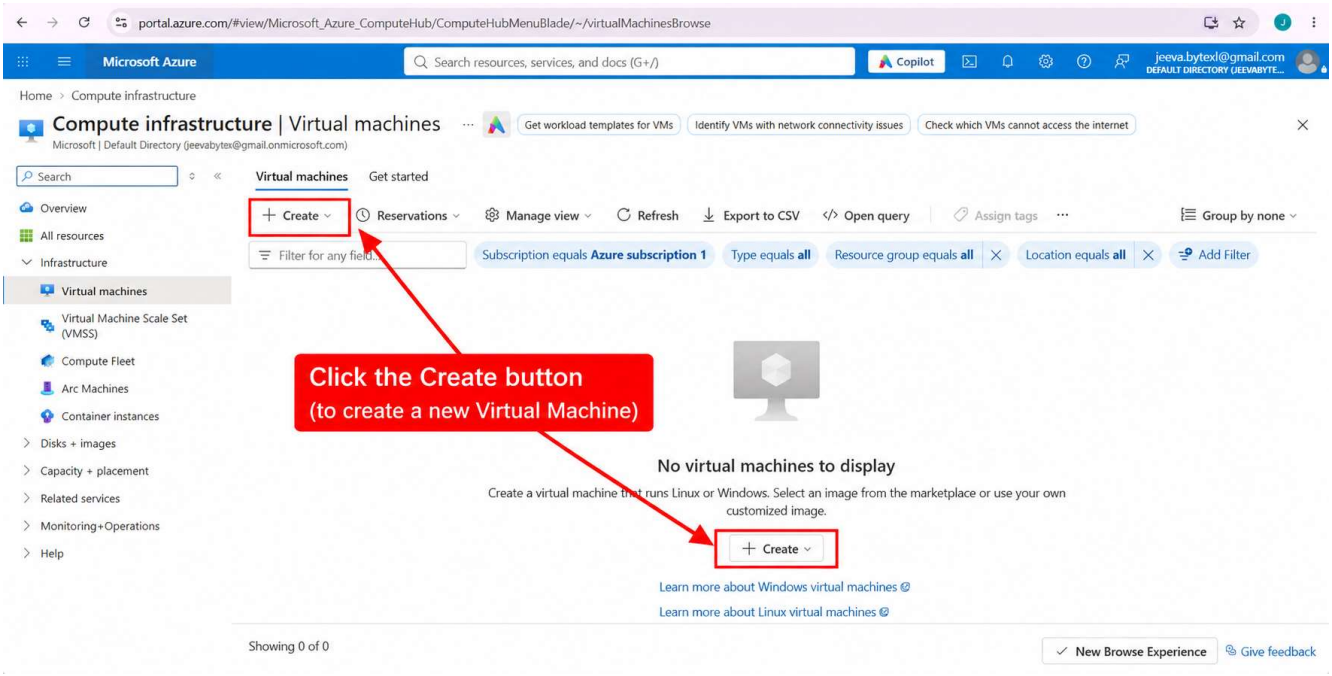
In the Azure portal, use the top search bar and type “virtual machine”. From the Services results, select “Virtual machines”. Do not select Virtual Machine scale sets or a Marketplace item.



Next: On the Virtual machines page, select Create, then choose Azure virtual machine to begin configuring your VM.

Step 12: Click Create to Start a Virtual Machine

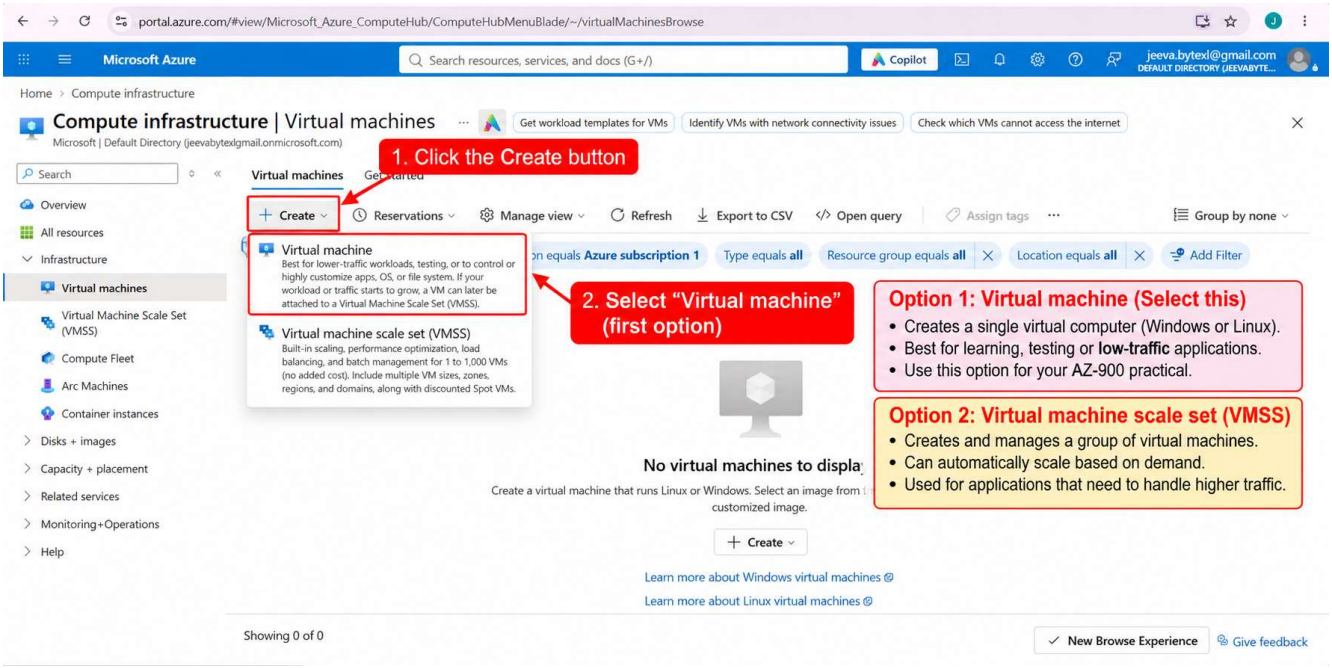
On the Virtual machines page, click the + Create button. The portal may show a Create menu; choose Azure virtual machine to start configuring a new virtual machine.



Note: The page may show a Create button near the top and another in the empty-state message. Either can open the creation menu.

Step 13: Select a Single Virtual Machine

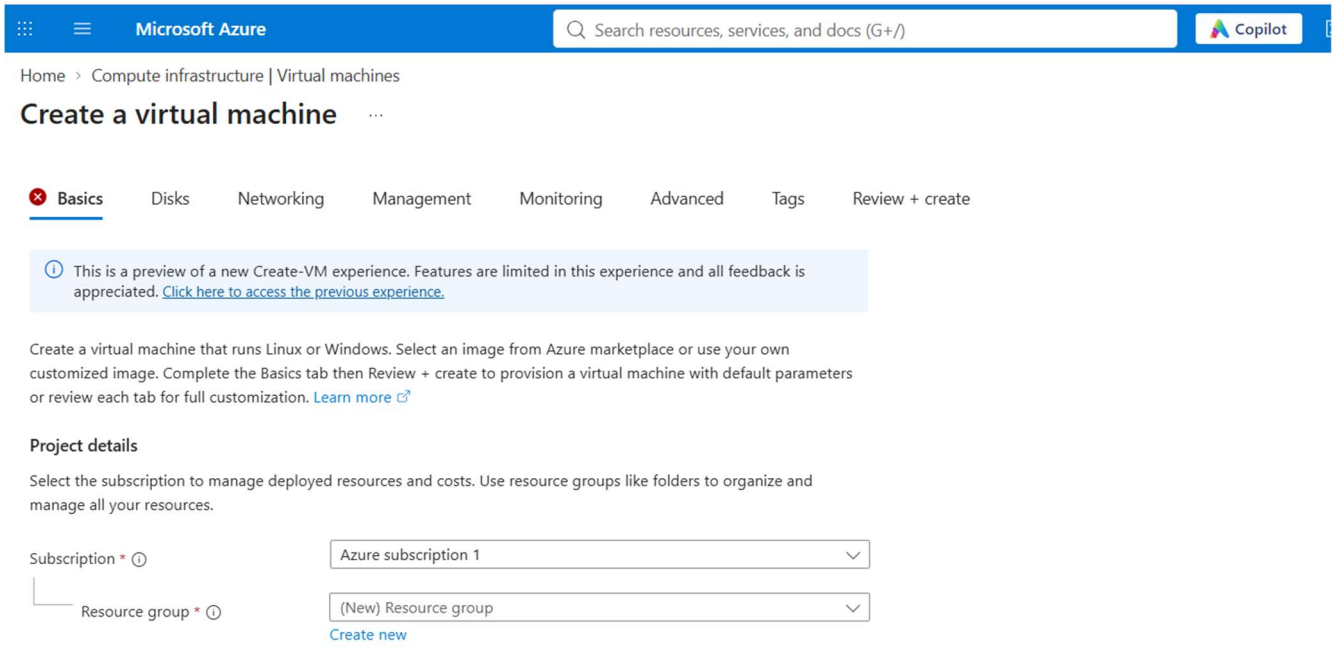
After clicking Create, choose the first option: Virtual machine. This creates one virtual computer running Windows or Linux and is the correct choice for this beginner AZ-900 practical.



The second option, Virtual machine scale set (VMSS), creates and manages a group of virtual machines that can scale based on demand. Skip this option for the current lab.

Step 14: Create a Virtual Machine — Basics (Project Details)

The Basics tab is the first page of the Virtual Machine creation form. Start by checking the subscription and selecting the resource group where the VM will be organized.



1. Subscription

The subscription manages Azure resource usage, billing, and access. Select the subscription you intend to use for the lab. In this screenshot, “Azure subscription 1” is selected.

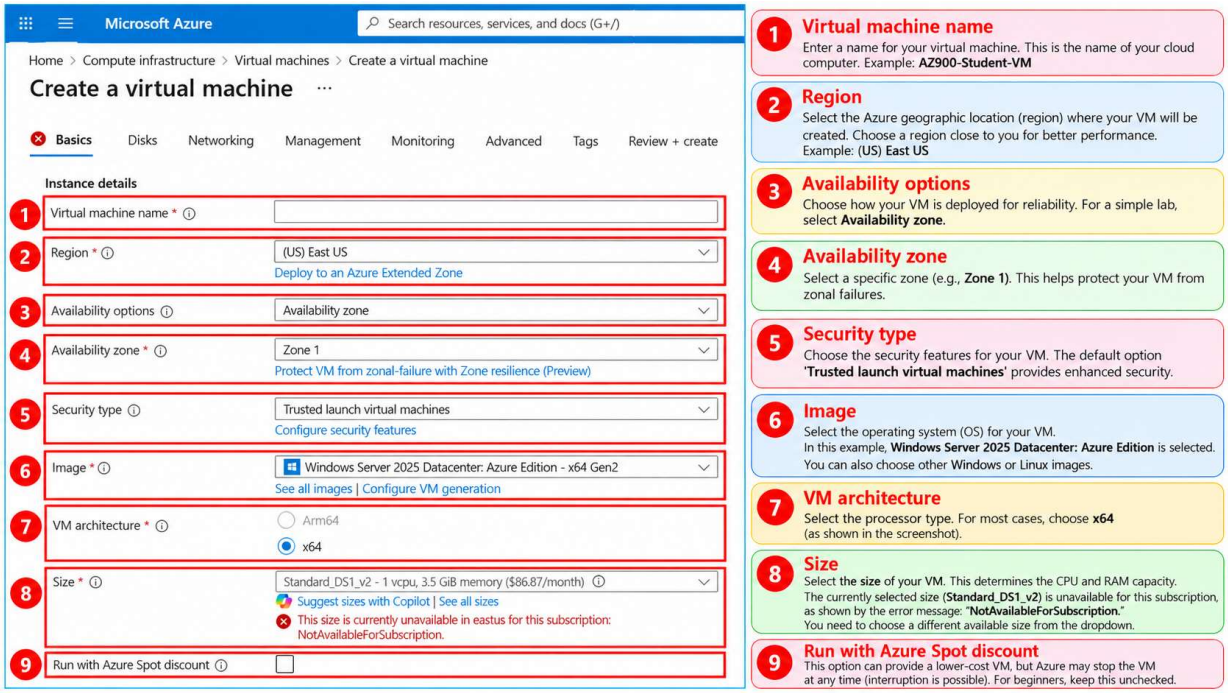
2. Resource group

A resource group is a logical container (like a folder) that keeps related Azure resources together. Choose the resource group you created earlier from the dropdown. If you have not created one, select “Create new” and give it a clear name, such as AZ900-Lab-RG.

For this practical, use the same subscription and your lab resource group, then continue down the Basics tab to configure the virtual machine name, region, operating system image, size, and administrator login.

Create a Virtual Machine – Basics: Instance Details

On this part of the Basics tab, configure the VM name, location, operating system, processor architecture, and size. These settings determine what computer Azure creates and where it runs.



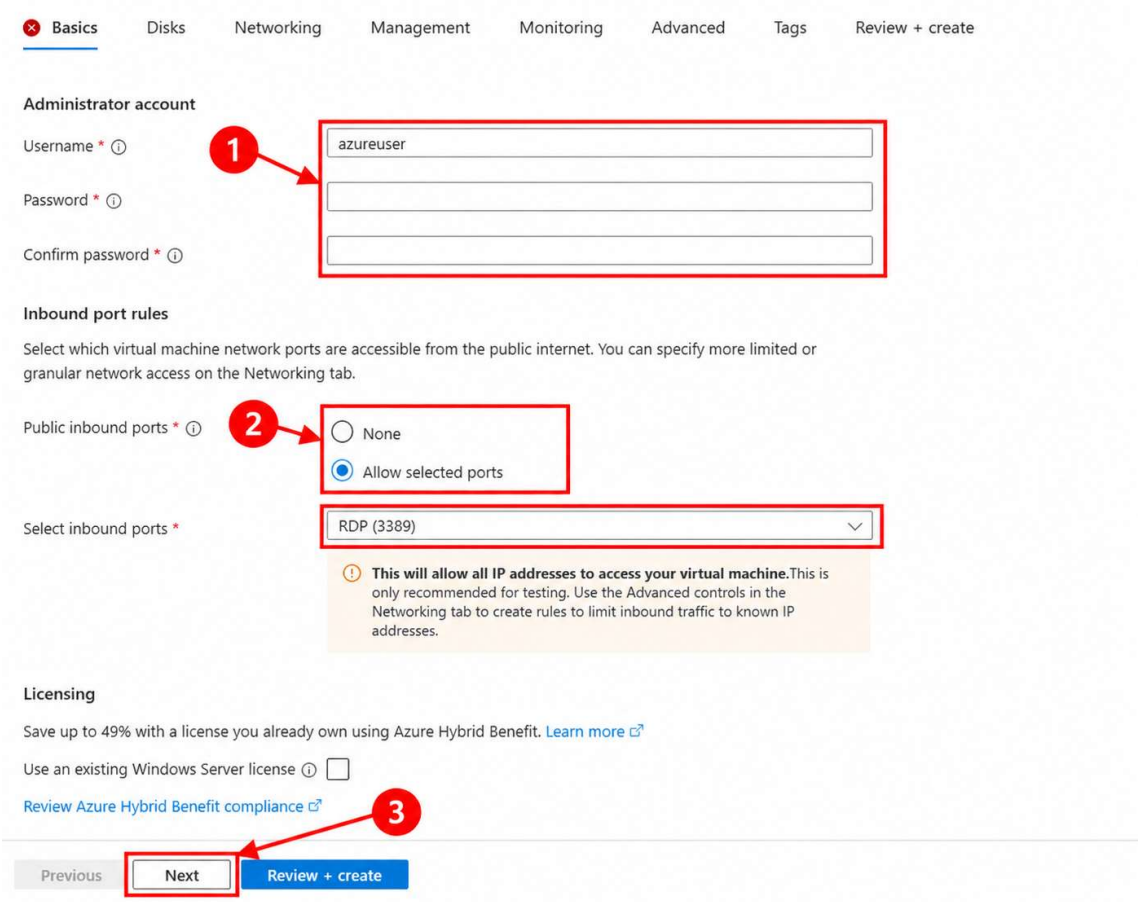
What each setting means

- **1. Virtual machine name:** The name used to identify your cloud computer. Example: AZ900-Student-VM.
- **2. Region:** The Azure geographic location where the VM is hosted. Choose a region allowed by your subscription and suitable for your lab.
- **3. Availability options:** Controls how the VM is placed for resilience. For a simple practice VM, use the default or the option your instructor specifies.
- **4. Availability zone:** A separate location within a region that can help protect against a zone-level failure. It may not be available for every VM size or region.
- **5. Security type:** Chooses the VM security configuration. Trusted launch provides additional protections when supported by the selected image and size.
- **6. Image:** The operating-system template used to create the VM, such as Windows Server or Linux.
- **7. VM architecture:** The processor architecture. x64 is a common choice for typical Windows and Linux lab VMs; select a compatible option for the chosen image.
- **8. Size:** Defines the virtual CPU, memory, and performance capacity. If Azure displays NotAvailableForSubscription, select another size available to your subscription.
- **9. Run with Azure Spot discount:** Uses spare Azure capacity at a lower price, but Azure can interrupt the VM. Leave this unchecked for a beginner lab unless specifically required.

Classroom tip: Explain the required fields first, and do not continue until the VM name, region, image, architecture, and size are valid. In the screenshot, the selected size is unavailable for the current subscription, so choose an available size before proceeding.

Step 15: Administrator Account and Inbound Port Rules

This part of the Basics tab sets the login credentials for the virtual machine and controls which network ports may be reachable from the public internet. Review these settings carefully before continuing.



1. Administrator account

Username: choose an administrator username. Password: create a strong password that meets Azure's requirements, and enter the same password in Confirm password. Keep the credentials private and store them securely. Do not use the example username or password as a shared class credential.

2. Public inbound ports

Choose None if you do not need public remote access during setup, or Allow selected ports when a connection is required. For Windows, Remote Desktop Protocol (RDP) commonly uses TCP port 3389. For Linux, Secure Shell (SSH) commonly uses TCP port 22.

Security note: Allowing a port from all IP addresses exposes the VM to the internet. Restrict remote access to trusted IP addresses in Networking/NSG rules where possible. Use only the access needed for the lab.

3. Licensing

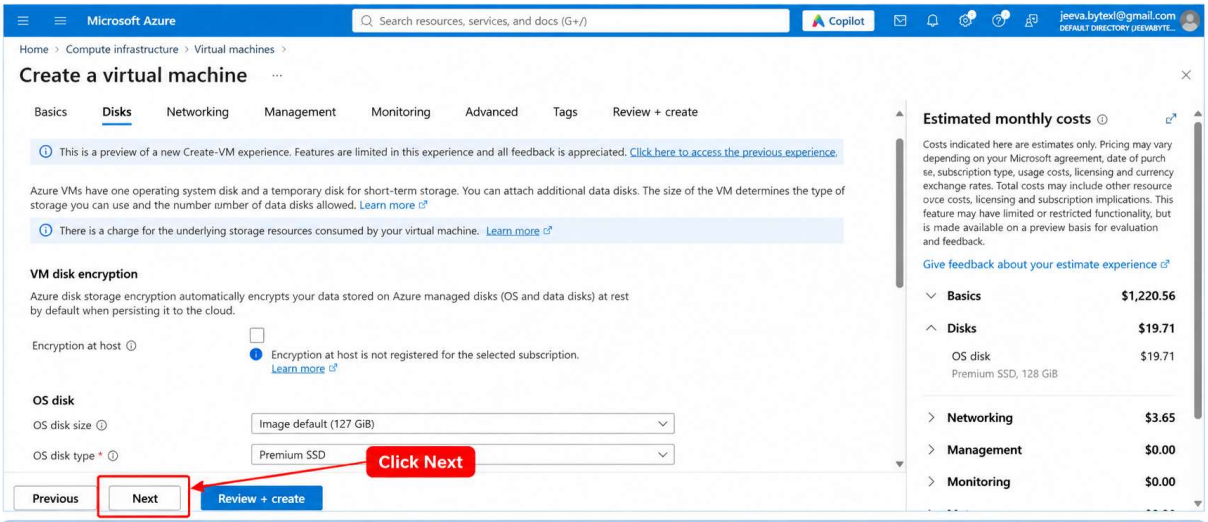
The Azure Hybrid Benefit checkbox is for eligible users who already have qualifying Windows Server licences. Leave it unchecked unless you know that you are eligible and authorised to use an existing licence.

4. Click Next

After the required username, password, confirmation, and inbound-port choices are valid, click Next to continue to the Disks section. For a beginner lab, review the settings before moving on.

Step 16: Configure Virtual Machine Disks

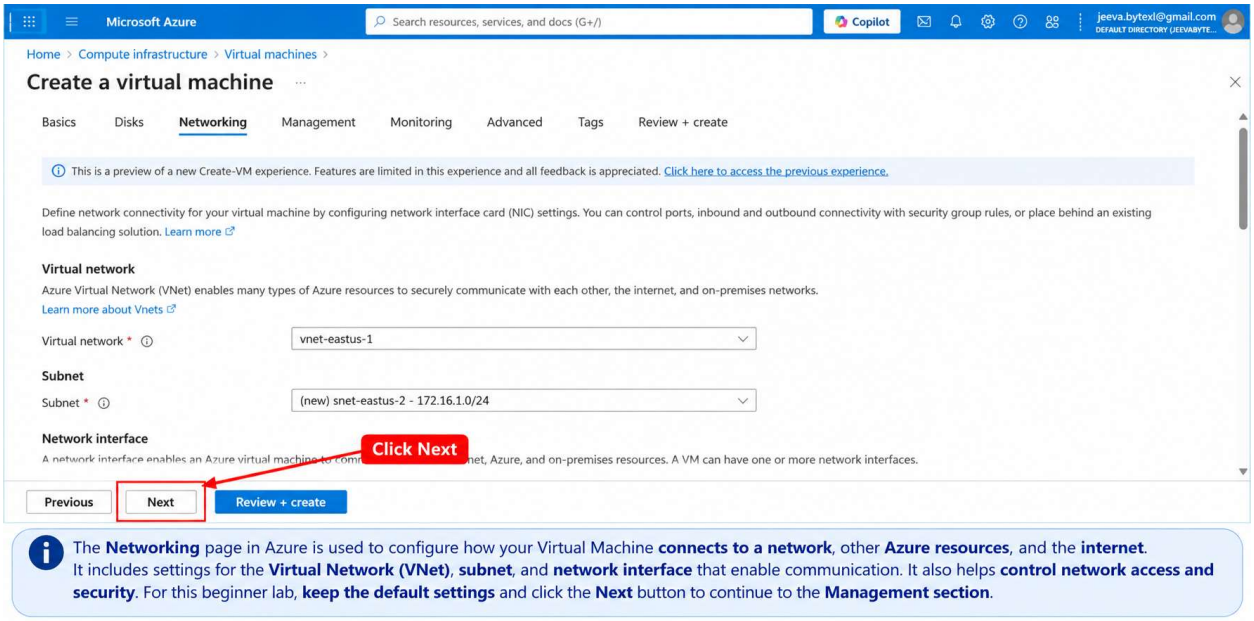
The Disks page in Azure is used to configure storage for your Virtual Machine. It determines where the operating system and other data are stored, the storage capacity and performance, and how the disks are managed. Azure also displays the estimated storage cost on this page. For this beginner lab, keep the default settings, review the estimated cost, and click Next to continue to Networking.



The **Disks** page in Azure is used to configure the **storage for your Virtual Machine.** It determines where the **operating system** and **other data** are stored, the **storage capacity and performance,** and how the **disks are protected and managed.** Azure also displays the **estimated storage cost** on this page. For a beginner lab, **keep the default settings, check the estimated cost,** and click the **Next** button to continue to Networking.

Step 17: Configure Networking

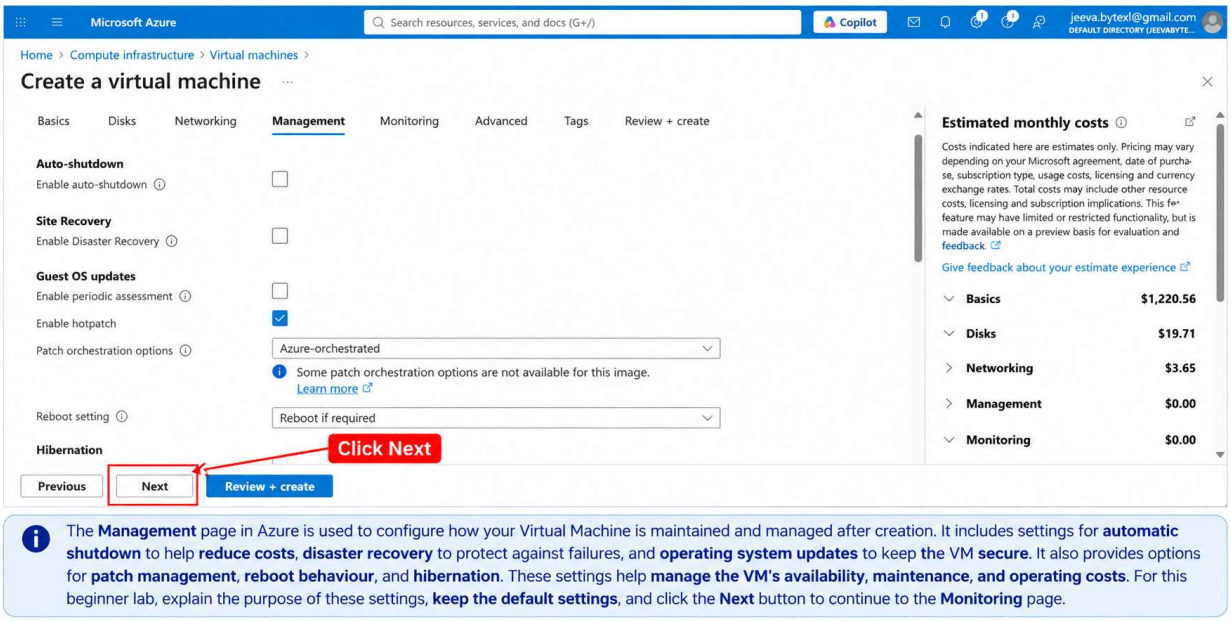
The Networking page in Azure is used to configure how your Virtual Machine connects to a network, other Azure resources, and the internet. It includes settings for the Virtual Network (VNet), subnet, and network interface that enable communication. It also helps control network access and security. For this beginner lab, keep the default settings and click Next to continue to the Management section.



The **Networking** page in Azure is used to configure how your Virtual Machine **connects to a network**, other **Azure resources**, and the **internet**. It includes settings for the **Virtual Network (VNet)**, **subnet**, and **network interface** that enable communication. It also helps **control network access and security**. For this beginner lab, **keep the default settings** and click the **Next** button to continue to the **Management section**.

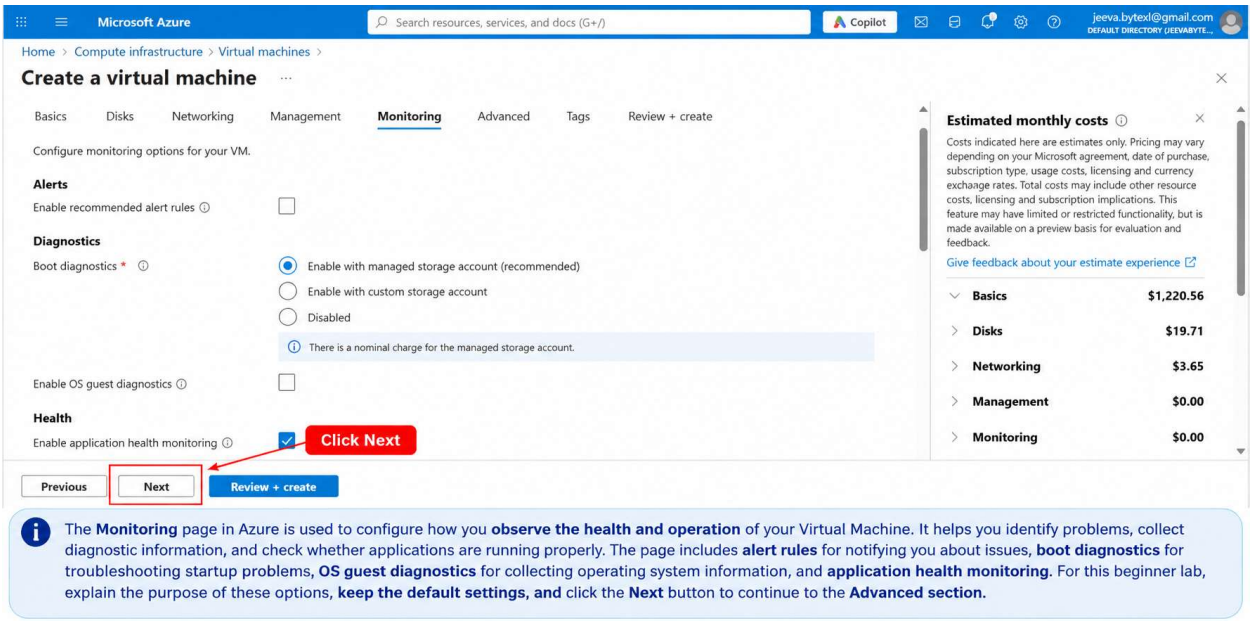
Step 18: Configure Management Settings

The Management page in Azure is used to configure how your Virtual Machine is maintained and managed after creation. It includes settings for automatic shutdown to help reduce costs, disaster recovery to protect against failures, and operating system updates to keep the VM secure. It also provides options for patch management, reboot behaviour, and hibernation. These settings help manage the VM's availability, maintenance, and operating costs. For this beginner lab, explain their purpose, keep the default settings, and click Next to continue to the Monitoring page.



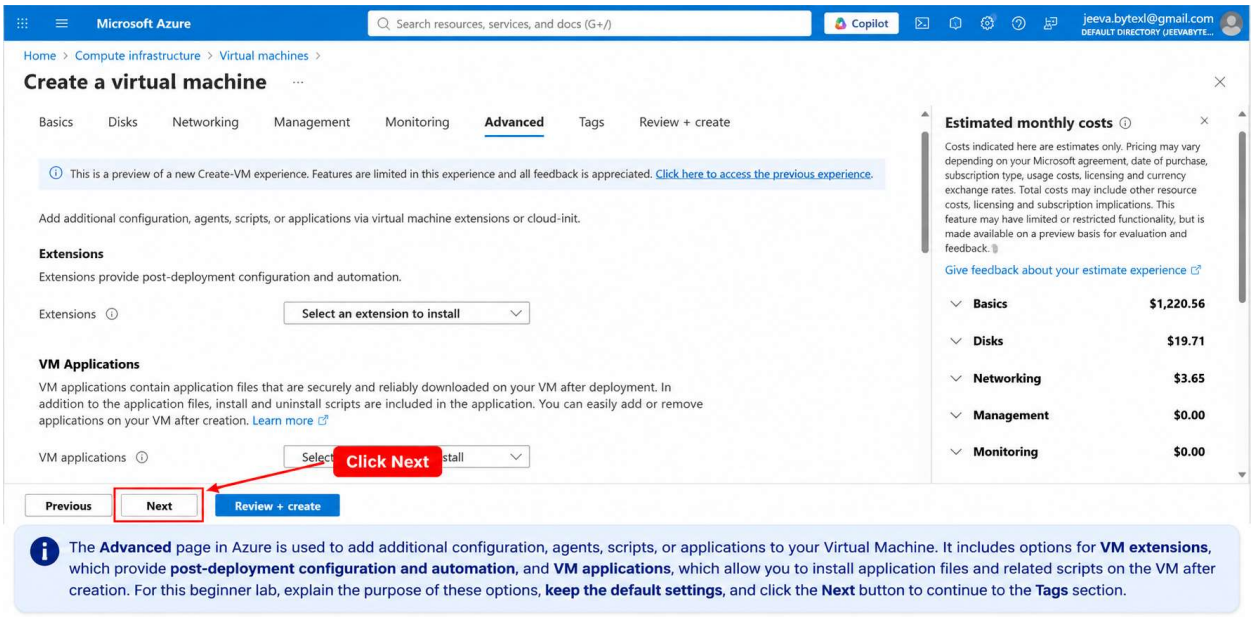
Step 19: Configure Monitoring Settings

The Monitoring page in Azure is used to configure how you observe the health and operation of your Virtual Machine. It helps you identify problems, collect diagnostic information, and check whether applications are running properly. The page includes alert rules for notifying you about issues, boot diagnostics for troubleshooting startup problems, OS guest diagnostics for collecting operating system information, and application health monitoring. For this beginner lab, explain the purpose of these options, keep the default settings, and click Next to continue to the Advanced section.



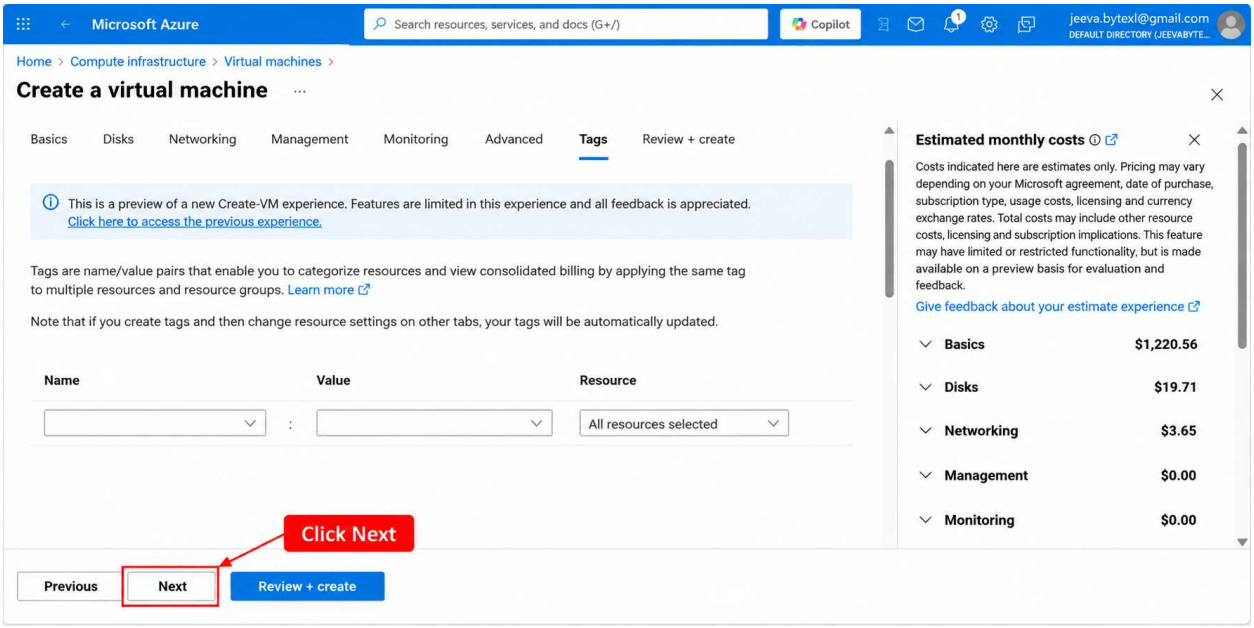
Step 20: Configure Advanced Settings

The Advanced page in Azure is used to add additional configuration, agents, scripts, or applications to your Virtual Machine. It includes options for VM extensions, which provide post-deployment configuration and automation, and VM applications, which allow you to install application files and related scripts on the VM after creation. For this beginner lab, explain the purpose of these options, keep the default settings, and click Next to continue to the Tags section.



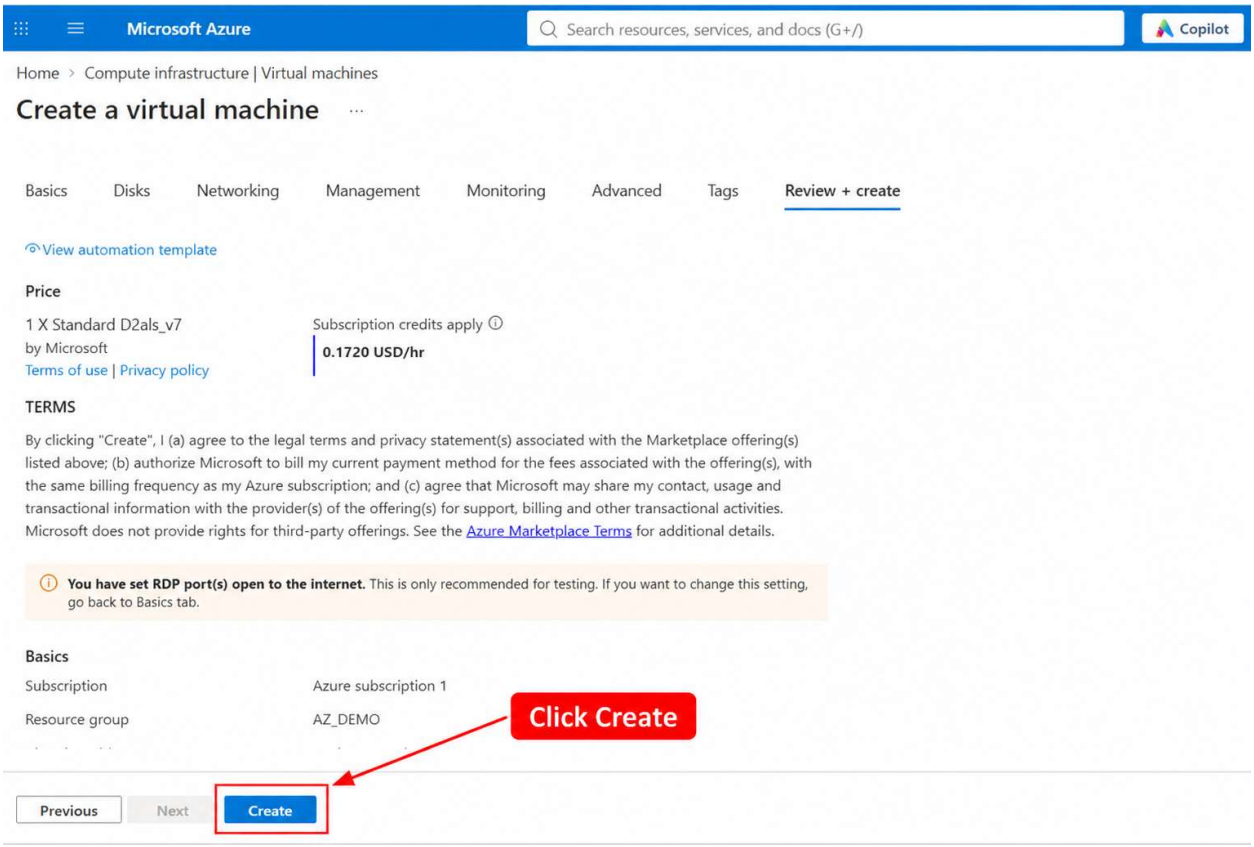
Step 21: Review the Tags Section

The Tags page lets you add labels to Azure resources using name-and-value pairs. Tags help categorize resources, identify their purpose or owner, and organize cost and billing reports. For this beginner lab, tags are optional, so leave the fields blank and click Next or Review + create to continue.



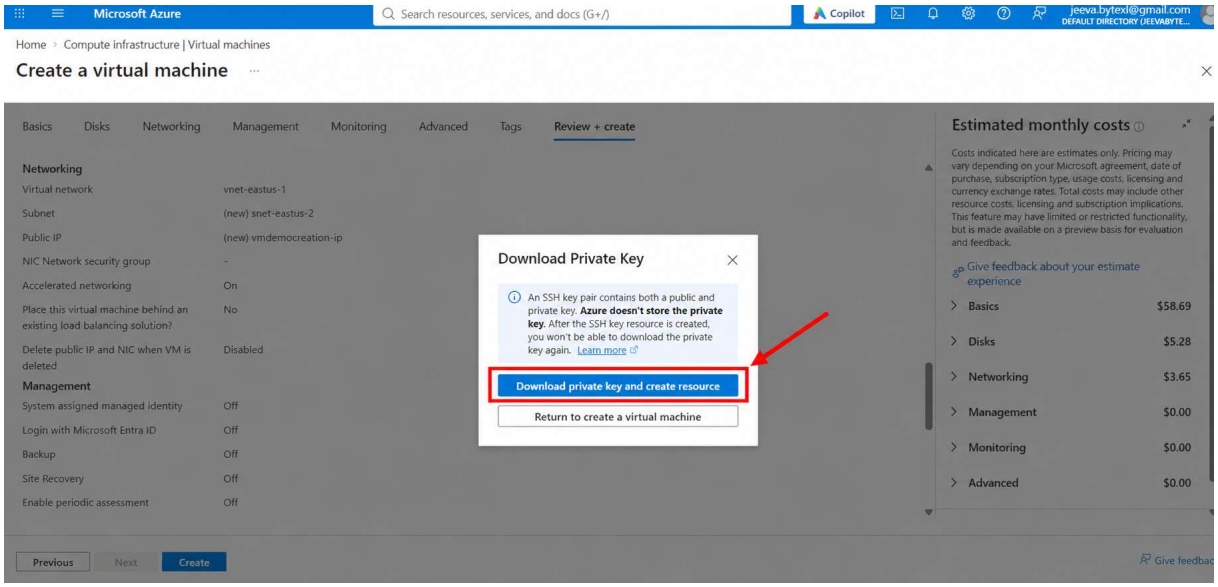
Step 22: Review and Create the Virtual Machine

The Review + create page shows a summary of the Virtual Machine settings you selected, along with the estimated price and applicable terms. Check the configuration and cost carefully before proceeding. Azure may display security warnings, such as an RDP port open to the internet; for a real deployment, restrict access to trusted IP addresses where possible. When the settings are correct, click Create to start deploying the VM.

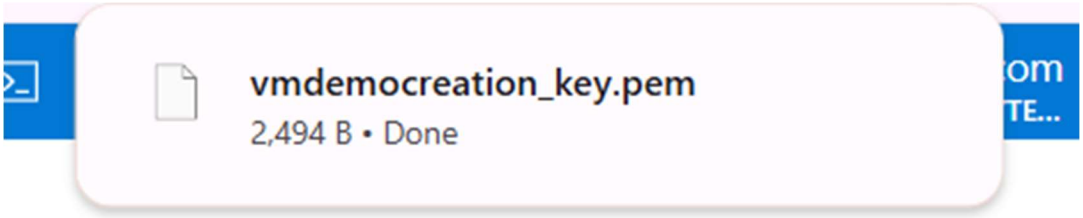


Step 23: Download the SSH Private Key

On the Review + create page, Azure may prompt you to download the SSH private key. Select “Download private key and create resource” to save the .pem file and continue creating the Linux virtual machine.

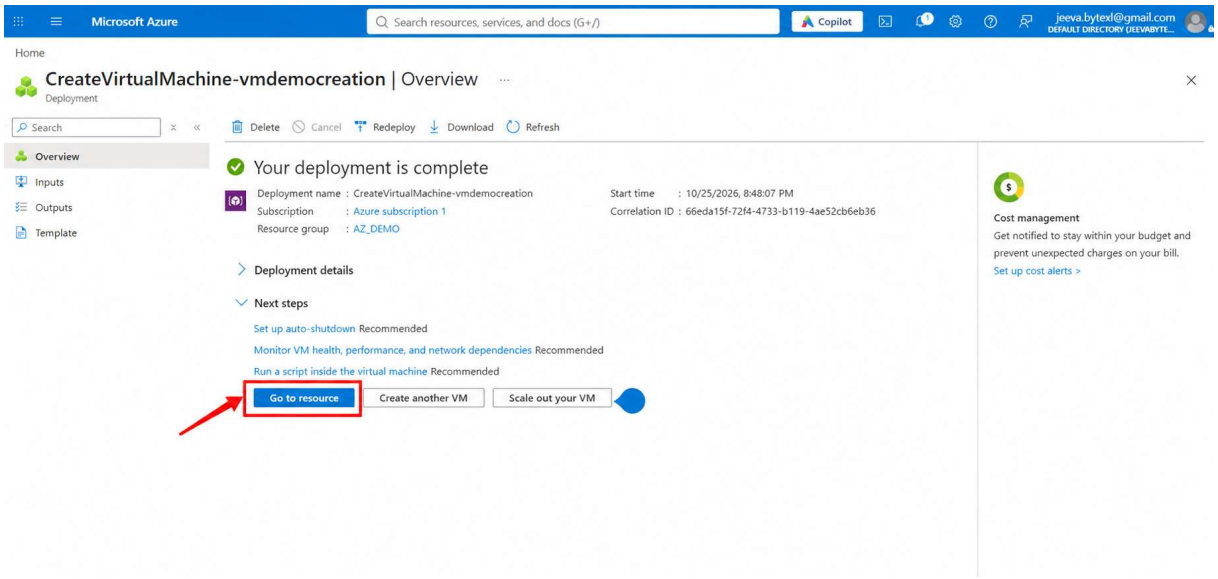


After the download completes, check your browser’s downloads. Keep the .pem private key in a safe place; Azure does not let you download that private key again. You will need it to connect to the Linux VM over SSH.



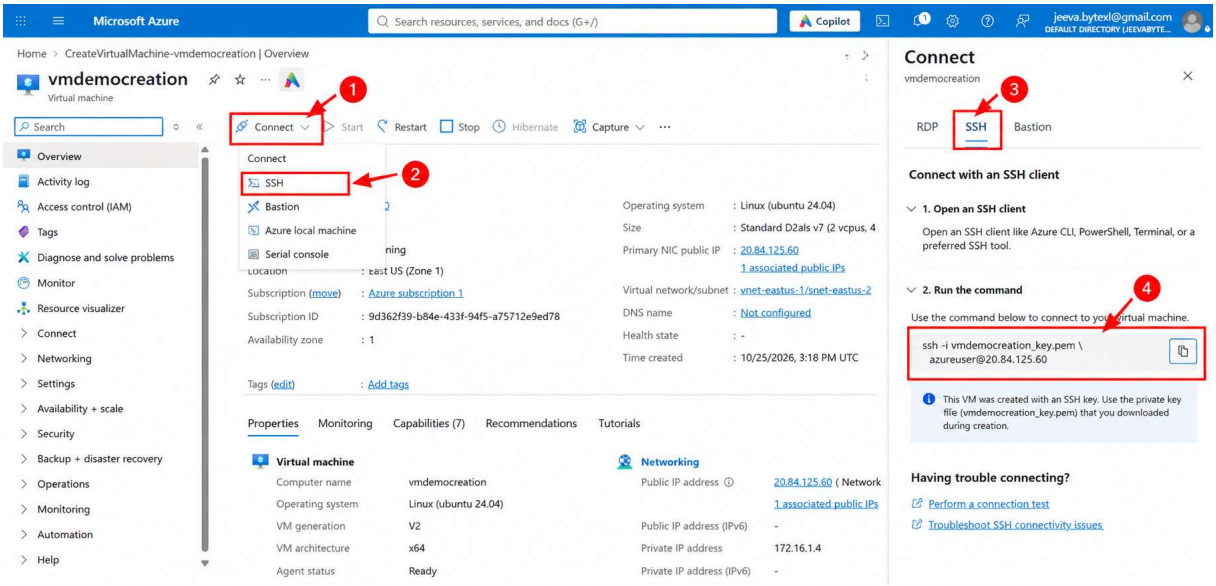
Step 24: Open the Created Virtual Machine

After Azure reports that deployment is complete, select “Go to resource” to open the newly created virtual machine.


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
jeeva.bytexl@gmail.com
DEFAULT DIRECTORY (JEEVABYTE...
Home
CreateVirtualMachine-vmdemocreation | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name : CreateVirtualMachine-vmdemocreation
Subscription : Azure subscription 1
Resource group : AZ_DEMO
Start time : 10/25/2026, 8:48:07 PM
Correlation ID : 66eda15f-72f4-4733-b119-4ae52cb6eb36
Deployment details
Next steps
Set up auto-shutdown Recommended
Monitor VM health, performance, and network dependencies Recommended
Run a script inside the virtual machine Recommended
Go to resource
Create another VM
Scale out your VM
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >

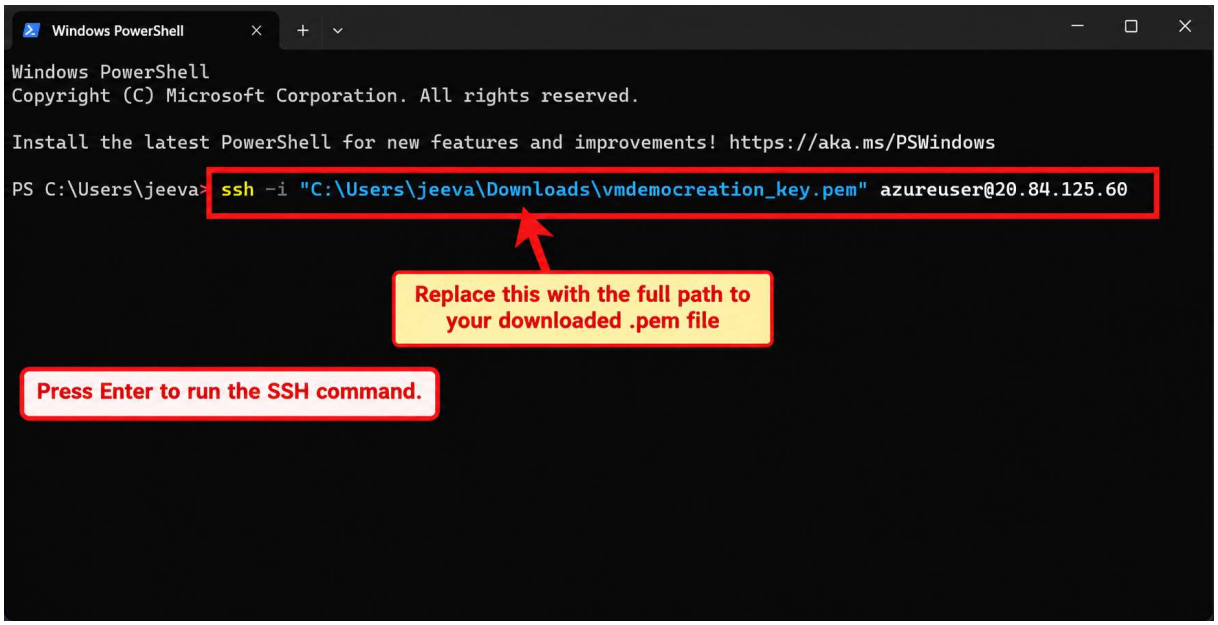
Step 25: Connect to the Linux VM Using SSH

On the virtual machine Overview page, select Connect, choose SSH, and follow the SSH client instructions shown by Azure. Use the downloaded .pem private key when running the connection command. Keep the private key secure.



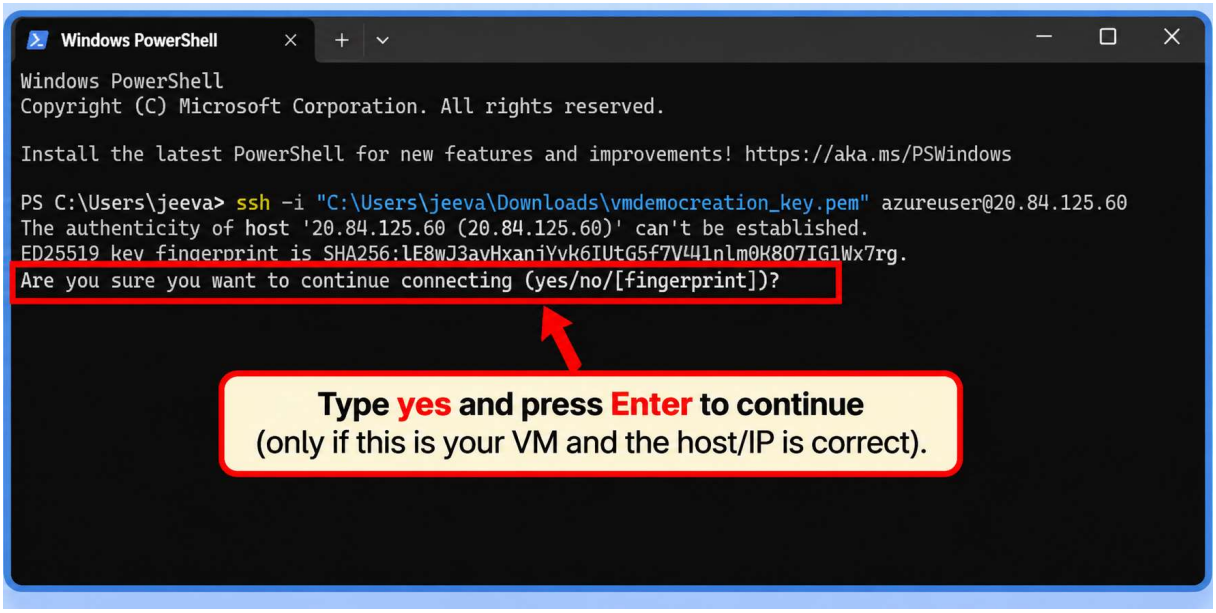
Step 26: Run the SSH Command in PowerShell

Open Windows PowerShell and paste the SSH command shown by Azure. Replace the example private-key path with the full path to your downloaded .pem file. Confirm the username and public IP address match your VM, then press Enter to connect.



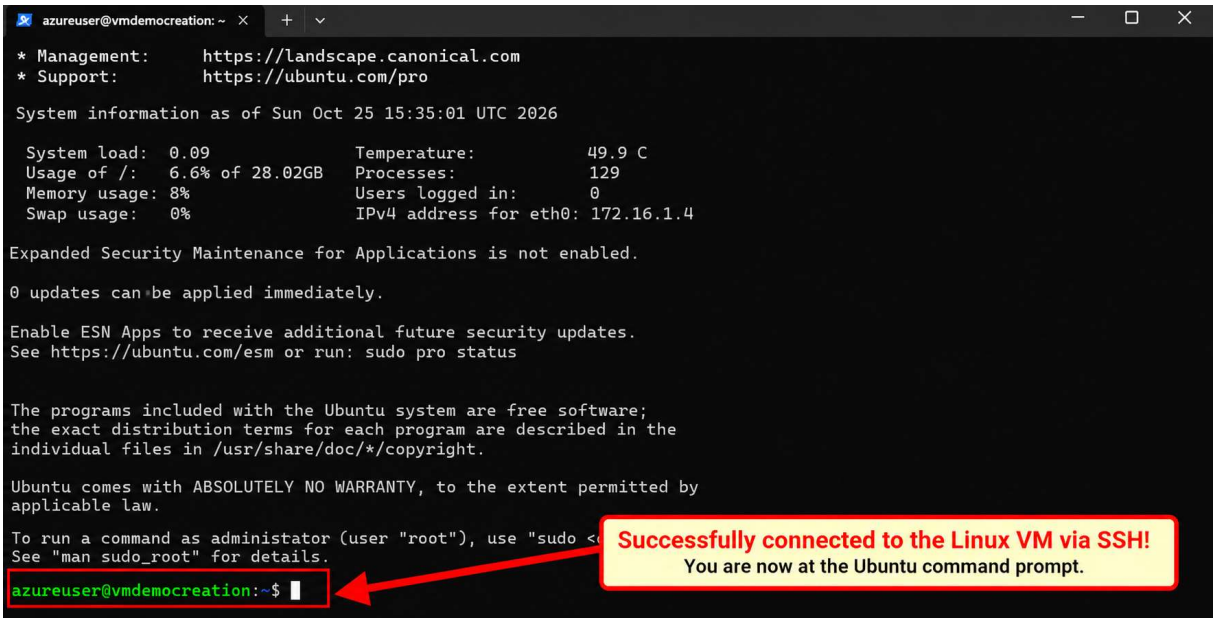
Step 27: Confirm the SSH Host

The first time you connect, PowerShell may ask whether you trust the host. If you have verified that the IP address belongs to your Azure VM, type yes and press Enter. Do not accept the prompt if the host or IP address is unexpected.



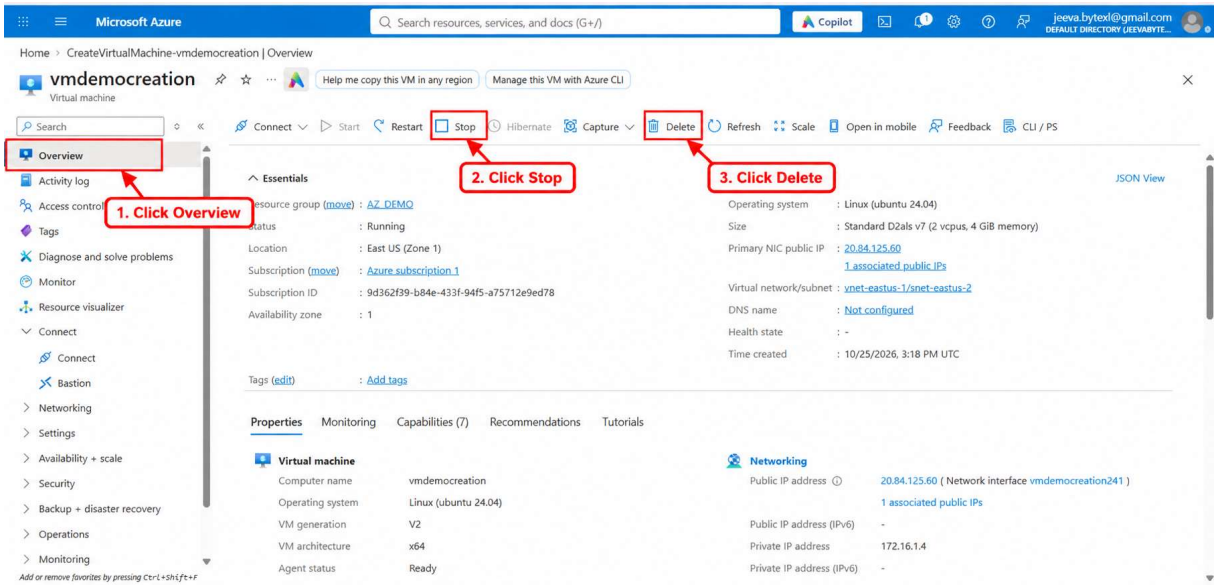
Step 28: Verify the Successful SSH Connection

When the SSH connection succeeds, the Ubuntu terminal displays system information and a command prompt such as azureuser@vmdemocreation:~$. This confirms that you are connected to the Linux virtual machine and can run Linux commands.



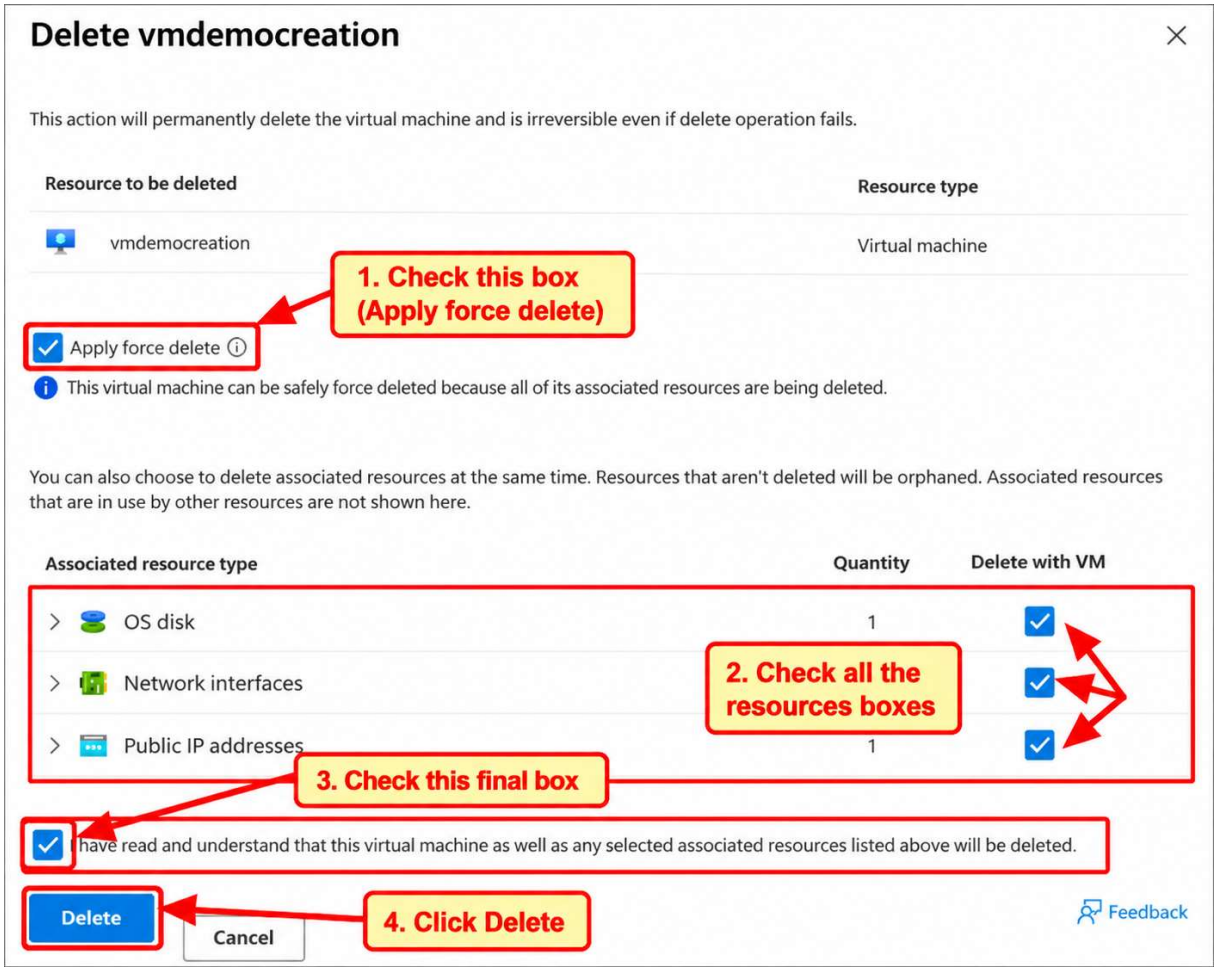
Step 29: Stop and Delete the Virtual Machine

After you finish the lab, stop and delete the virtual machine if you no longer need it. On the VM Overview page, select Stop and wait until the VM is stopped. Then select Delete and follow the confirmation prompts. Deletion is permanent for the VM resource. Review related resources such as disks and public IP addresses before confirming, because they may remain and continue to incur charges unless you remove them too.



Step 30: Confirm VM and Associated Resource Deletion

In the Delete confirmation panel, review the listed resources. Select Apply force delete if appropriate, check the boxes for the associated resources you intend to delete, and select the final acknowledgement checkbox. Then click Delete to confirm. This permanently deletes the selected resources, so verify your choices before proceeding.



Step 31: Refresh and Verify the VM Was Deleted

After confirming the deletion, return to the Azure portal and refresh the page. The virtual machine should no longer appear in the VM list. If it still appears briefly, wait for the deletion operation to finish and refresh again.

Final check: open the resource group and review its remaining resources. Delete any associated resources you no longer need, such as disks, network interfaces, or public IP addresses, if they were not deleted with the VM. Only remove resources you are sure are not used by another workload.